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Behind The Picture

Ever wondered what makes graphics cards tick, and tick so well?

Michael Browne

From the movies we watch to medical sciences and imaging, educational simulations to defence applications, and even manufacturing processes (Computer Assisted Manufacturing) the impact that 3D rendering has had on the world is phenomenal. 3D accelerators (or graphics cards) have acted as catalysts for the growth of 3D modelling, rendering, and animation.

The year 1981 and the first PC video card from IBM seems ancient history. We've gone from single colour video cards to cards that can work with millions of colours. The real graphics war for desktop PCs didn't start till 1997 when 3dfx delivered *Voodoo* to the world with (then) powerful features like Mip-Mapping, Z-buffering and AntiAliasing. Rivals NVIDIA came up with their TNT and TNT2 cards following the Voodoo 2, and these were the first AGP-based cards. Though it may seem that things have stagnated from those initial gigantic steps, nothing could be further from the truth. Enter NVIDIA's GeForce 256 graphics card (August 31, 1999). It essentially had 4 pixel pipelines that could each handle a single, half precision (16 bit) pixel operation in a clock. Core and memory speeds were 120/166 MHz and with a peak output of 480 million pixels per second and 15 million triangles per second, the GeForce 256 was hard to beat. November 8, 2006, seven scant years later and the GeForce family's eighth generation of graphics cards were unveiled. Their latest 8800GT, which is a very value oriented card, churns out 16.8 billion triangles per second, a difference comparable to the difference between the sun and the moon. Regardless of the difference in specifications, there are certain characteristics common to all graphics cards which we're going to look at. We're also going to see how all this comes together. Let's look at the individual parts of a graphics card, and what they contribute to the whole.

